Code

assignment5.py > ...

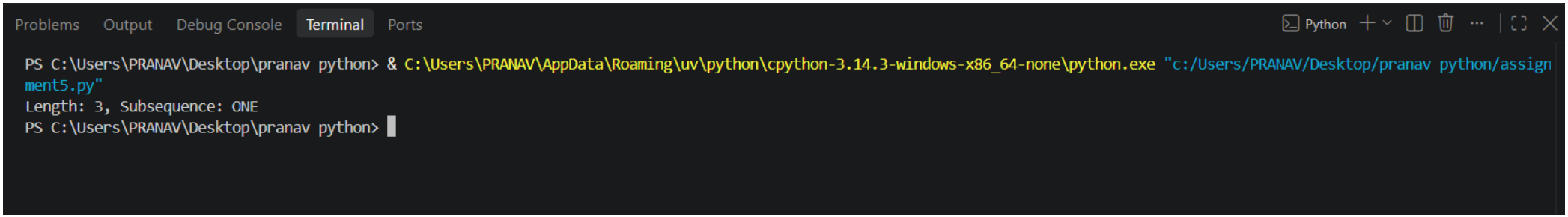
```python
def lcs(X: str, Y: str):
    m, n = len(X), len(Y)
    dp = [[0] * (n + 1) for _ in range(m + 1)]

    for i in range(1, m + 1):
        for j in range(1, n + 1):
            if X[i - 1] == Y[j - 1]:
                dp[i][j] = dp[i - 1][j - 1] + 1
            else:
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])
    lcs_chars = []
    i, j = m, n
    while i > 0 and j > 0:
        if X[i - 1] == Y[j - 1]:
            lcs_chars.append(X[i - 1])
            i -= 1
            j -= 1
        elif dp[i - 1][j] >= dp[i][j - 1]:
            i -= 1
        else:
            j -= 1

    return dp[m][n], "".join(reversed(lcs_chars))

def assignment5():
    length, subsequence = lcs("STONE", "LONGEST")
    print(f"Length: {length}, Subsequence: {subsequence}")
assignment5()
```

Output


Problems
Output
Debug Console
Terminal
Ports
Python
PS C:\Users\PRANAV\Desktop\pranav python> & C:\Users\PRANAV\AppData\Roaming\uv\python\cpython-3.14.3-windows-x86_64-none\python.exe "c:/Users/PRANAV/Desktop/pranav python/assignment5.py"
Length: 3, Subsequence: ONE
PS C:\Users\PRANAV\Desktop\pranav python>